Organic solar cells reach 21% efficiency with two-step crystallization process

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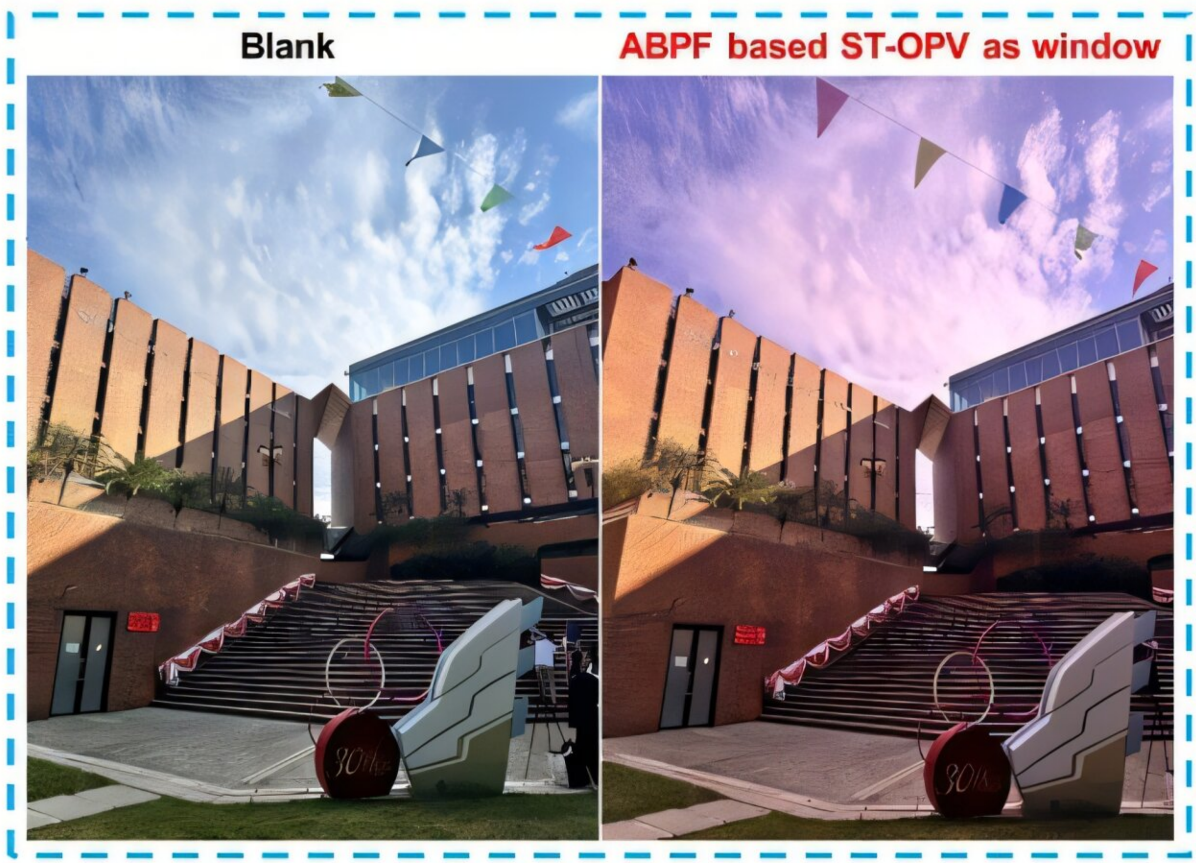


Transparent organic solar cells achieved record light utilization efficiency of 6.05%. Credit: Fu et al. "Semitransparent organic photovoltaics with wide geographical adaptability as sustainable smart windows," 2025 Nature Communications.

While most solar cells on the market today are based on silicon, energy engineers have recently been assessing the performance of alternative cells based on other photovoltaic (PV) materials. These alternative options include so-called organic solar cells (OSCs), lightweight and flexible cells that are based on organic semiconducting materials.

The operation of OSCs relies on a so-called active layer, a structure made of two different types of materials, referred to as donor and acceptor materials. Both materials absorb sunlight and generate excitons which dissociate into electrons and holes at the interface between donor and acceptor materials. Then holes are transported in donor materials, while the acceptors transport electrons and facilitate their flow through the device to generate electricity.

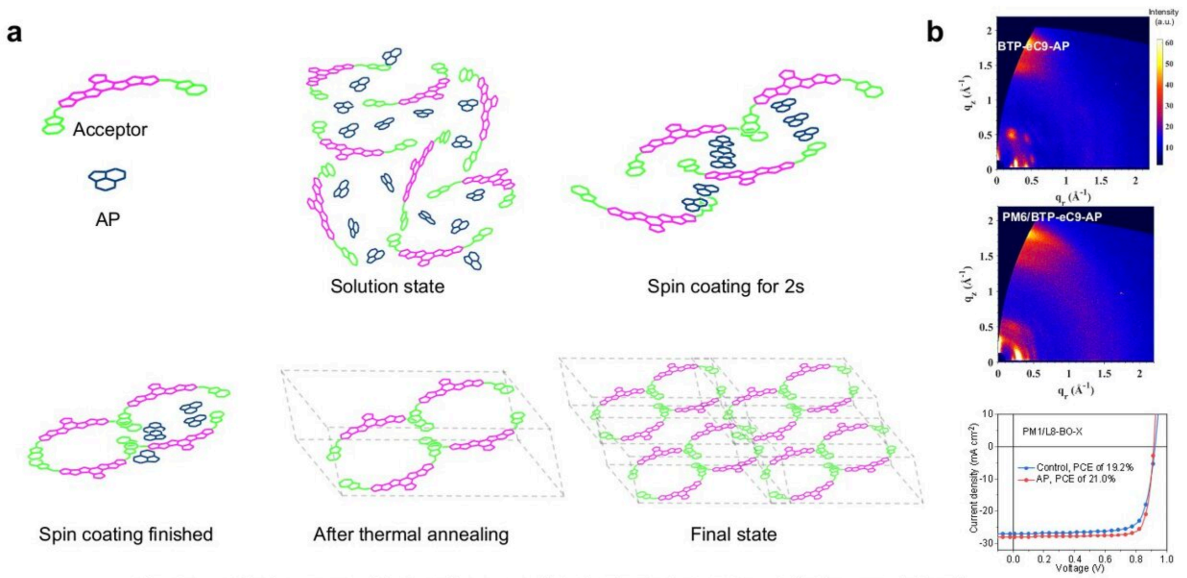
Compared to conventional silicon-based solar cells, OSCs could be more flexible, lighter, more affordable and easier to tailor for specific applications, for instance by changing their color or transparency. Nonetheless, the efficiency with which they convert solar energy into electricity remains significantly lower than that of commercially available photovoltaics (PVs).

A promising strategy to increase the efficiency of OSCs is to utilize a class of acceptor materials known as non-fullerene acceptors (NFAs). Unfortunately, the organic molecules in these materials have proved to be difficult to crystallize into the uniform lattices that would enable desired efficiency gains.

Researchers at the Hong Kong Polytechnic University, Sichuan University, and other institutes recently introduced a new two-step crystallization process that could yield more uniform NFAs. Using this approach, outlined in a paper published in *Nature Energy*, they were able to attain high-quality acceptor materials that significantly improved the efficiency of OSCs.

"OSCs active layer comprising a blend of donor (D) and acceptor (A) materials, and the nanoscale arrangement of the donor and acceptor molecules in the active layer (morphology) is key to increasing device efficiency, yet it remains challenging," Prof. Gang Li, senior author of the paper, told Tech Xplore.

"In this work, we introduced acenaphthene (AP) as a crystallization-regulating agent to manipulate the crystallization dynamics of NFAs. The resulting OSC active layer shows unprecedented highly oriented stacking of the NFA molecules, with well-defined Bragg spots. We achieved a PCE of 21% (20.5% certified) with a maximum FF of 83.2% (82.2% certified)."



Two-step crystallization process and device performance. a, Illustration showing the two-step crystallization process induced by acenaphthene. The active layer is formed through spin coating of the photoactive molecules with acenaphthene (AP)-, followed by thermal annealing. b, Two-dimensional grazing incident wide-angle X-ray scattering (GIWAXS) diffraction pattern of the control BTP-eC9 film (top) and the acenaphthene (AP)-treated BTP-eC9 film (bottom). The Bragg diffraction spots in the AP-treated film indicate highly ordered molecular packing. c, Current density–voltage curves of PM1/L8-BO-X OSCs with and without AP processing.

Credit: Fu et al.

Two decades ago, Prof. Li published a paper in *Nature Materials*, where he reported a new approach to improve the morphology of OSCs. In this paper, he showed that the self-organization of polymer molecules could significantly increase the crystallinity of the polymer–fullerene active layer in OSCs, boosting their efficiency.

Building on this discovery, he set out to identify promising strategies that would improve the crystallization of the active layer in OSCs. This ultimately led to his most recent study, conducted in collaboration with researchers working at various institutes across China.

"In this work, we first used the layer-by-layer method to deposit the donor and acceptor and form bi-continuous phases, to achieve the ideal vertical phase separation in the active layer," explained Prof. Li. "The crystallization-regulating agent—acenaphthene is a key to manipulating the crystallization dynamics of the NFA layer on top of the polymer donor film, resulting in quasi-single crystalline level active layer ordering."

To assess the potential of their newly proposed crystallization process, the researchers monitored the formation of thin film organic acceptors using various techniques. In addition, they relied on a computational quantum mechanics-based method (i.e., performing functional theory calculations) to simulate the interactions between molecules and the regulator agent they used.

"We confirmed that acenaphthene induces a two-step crystallization with sustained delayed crystallization which enables high quality crystalline NFA and optimized OSC morphology," said Prof. Li. "This strategy is further proven to be very effective in several state-of-the-art OSC material systems."

The regulating agent-assisted deposition strategy employed by the researchers enabled the realization of uniform NFAs, which were then integrated in binary OSCs. These solar cells were found to achieve efficiencies up to 21%, which are remarkable for OSCs, but still not as high as those attained by silicon solar cells.

"High efficiency and stability are key requirements for new solar cells like OSC," added Prof. Li. "Upon satisfaction of these requirements, OSC technology could enable better and new solar applications including flexible, lightweight, portable, aesthetic, customizable semi-transparent solar applications.

"We will now continue to improve OSC performance—efficiency and stability. We will also work on nice form factor OSCs (flexible, aesthetically pleasing, transparent etc.), as well as scalable fabrication techniques, and attempt the cells' integration with other types of devices towards real-world applications such as smart power windows."

More information: Jiehao Fu et al, Two-step crystallization modulated through acenaphthene enabling 21% binary organic solar cells and 83.2% fill factor, *Nature Energy* (2025). DOI: 10.1038/s41560-025-01862-1.

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